Q. One: The mean of the process $X(t) = \cos(wt)$; $w \sim U(0, b)$ equals $t^{-1}\sin(t)\cos(t)$. The value of the positive constant b is equal to

A) 0.5 B) 1 C) 2 D) 3 E) None

Q. Two: For a random process $X(t)$ with $E(X^2(t)) = 3 + \cos(t) + \sin(\pi t)$ we have $P_{XX} =$

A) 0 B) 1 C) 2 D) 3 E) None

Q. Three: For two orthogonal random variables X, Y with $R_{XY} = 2\,E(Y)$, then $E(Y) =$

A) 0 B) 1 C) 2 D) 3 E) None

Q. Four: Orthogonal processes $X(t), Y(t)$ with $R_{XY}(\tau) = \bar{X}\,\bar{Y} - 1$, then $\bar{X}\,\bar{Y} =$

A) 0 B) 1 C) 2 D) 3 E) None

Q. Five: The mean and the variance of a mean-ergodic process $X(t)$ with nonzero mean, no periodic component and $R_{XX}(\tau) = \bar{X} + 4e^{-2|\tau|}$ are equal respectively to

A) 1 , 2 B) 2 , 1 C) 4 , 1 D) 1 , 4
E) None

Q. Six: A process $X(t)$ with $E(X(t)) > 0$, $C_{XX}(t, t+\tau) = E(X^2(t)) - \frac{t^2\cos(\tau)}{\tau+1}$.
We have $E(X(t)) =$

A) t^2 B) 1 C) t D) -1 E) None

Q. Seven: For $X \sim Poisson(m)$ with $P(X > 0) = 1 - e^{-2}$, we have $Var(X) =$

A) 2 B) 0.5 C) 1/3 D) 3
E) None

Q. Eight: For jointly wide-sense stationary discrete time random processes $X(nT_S), Y(nT_S)$ such that $C_{XY}(kT_S) = \cos(kT_S) - \bar{X}\,\bar{Y}$. Then $R_{XY}(kT_S) =$

A) $\bar{X}\,\bar{Y}$ B) $\cos(kT_S)$ C) $\cos(kT_S) - 2\bar{X}\,\bar{Y}$
D) $\sin(kT_S)$ E) None

Q. Nine: For $X \sim N(\sqrt{2}, 2)$ then $P(X > std(X)) =$

A) 0.25 B) 0 C) 0.5 D) 1 E) None

Q. Ten: A random variable X with pdf $f(x) = x,\ x \in (0, \sqrt{2})$. The pdf of $Y = \sqrt[3]{X}$ is

A) $3y^6$ B) $y^{1/3}$ C) $3y^{2/3}$ D) $3y^5$ E) None

Q. Eleven: Jointly wide-sense stationary orthogonal processes $X(t), Y(t)$ with $R_{XY}(\tau) = 2 + \bar{X}\bar{Y}$, then $C_{XY}(t, t+\tau) =$

A) 2 B) 3 C) 4 D) 5 E) None

Q. Twelve: which of the following if false

A) The null hypothesis is the one to be tested

B) If testing whether some parameter μ is different than 1, the hypotheses would be $H_0: \mu < 1$ and $H_A: \mu \geq 1$

C) A Type I error occurs when the null hypothesis is rejected, even though it is actually true

D) A Type II error occurs when the null hypothesis is not rejected, even though it is actually false

Q. Thirteen: Orthogonal processes $X(t), Y(t)$ with $R_{XY}(\tau) + \bar{X}\bar{Y} = 25$ and $\bar{X} + \bar{Y} = 10$. Then $\bar{X} =$

A) 0.5 B) 1 C) 5 D) 10 E) None

Q. Fourteen: If $Var(X) = 3$ and $Y = X - \bar{X}$, then the correlation of $2X$ and Y is

A) 2 B) 4 C) 8 D) 6 E) None

محمد ابو دية

Final ~~2024~~ 2025

1) $E(x(t)) = \frac{1}{t} \cdot \sin(t) \cdot \cos(t)$

$\cancel{\frac{1}{t}}\left(\sin(wt)\Big|_0^b\right) * \frac{1}{b} = \cancel{\frac{1}{t}} \cdot \sin(t) \cdot \cos(t)$

$\frac{\sin(bt)}{b} = \sin(t) \cdot \cos(t)$ → متطابقة $\sin(2x) = 2\sin(x)\cos(x)$

$\frac{\sin(bt)}{\frac{1}{2}\sin(2t)} = b \Rightarrow \boxed{b = 2}$ (C)

---

2) $P_{xx} = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} E(x^2(t)) \cdot dt$

$= \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} 3 + \cos(t) + \sin(\pi t)$ (D)

$= \lim_{T \to \infty} \frac{1}{2T}(6T + 0 + 0) \Rightarrow \lim_{T \to \infty} \frac{6\cancel{T}}{2\cancel{T}} = \boxed{3}$

---

3) oRthogonal $\Rightarrow R_{xy} = 0 \Rightarrow 2E(y) = 0 \Rightarrow$ (A) $E(y) = 0$

---

4) $R_{xy} = 0 = \bar{X}\bar{y} - 1 \Rightarrow \bar{X}\bar{y} = 1$ (B)

---

5) $R_{xx}(0) = E(x^2(t))$, $\lim_{\tau \to \infty} R_{xx}(\tau) = (E(x))^2$

$E(x) + 4 = E(x^2(t))$ $E(x) + 0 = (E(x))^2$

$4 = Var(x)$ $E(x) = 1$

1, 4 (D)

لجنة السنافر

6) $C_{xx}(t, t+\tau) = E(x^2(t)) - t^2 \cdot \frac{\cos(\tau)}{\tau + 1}$ find $E(x(t))$:

عمر ابو دية
Final 2025

$$\boxed{C_{xx}(t, t+\tau)\Big|_{\tau=0} = Var(\cancel{t}, \cancel{t}+\tau)}$$

$$\Rightarrow \cancel{E(x^2(t))} - (E(x(t)))^2 = \cancel{E(x^2(t))} - t^2$$

$$\boxed{E(x(t)) = t} \quad ⓒ$$

---

7) $\boxed{Var(x) = m}$ $\boxed{P(x > 0) = 1 - P(x < 0) = 1 - e^{-2}}$

$$\cancel{1} - \frac{e^{-m} \cdot m^x}{x!}\Big|_{x=0} = \cancel{1} - e^{-2}$$

Ⓐ $\boxed{m = 2}$ $\Leftarrow e^{-m} = e^{-2}$

~~m = 0~~

---

8) jointly wide-sense $\Rightarrow R_{xy}(KT_s) \Rightarrow R_{xy}(\tau)$

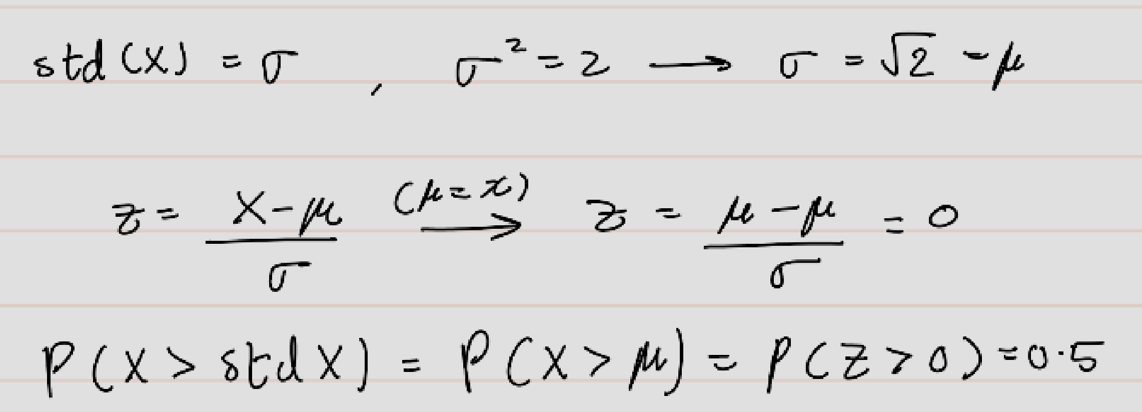
$C_{xy}(KT_s) = \cos(KT_s) - \bar{x}\,\bar{y}$

$\downarrow$

R_{xy} $\rightarrow$ $\boxed{R_{xy} = \cos(KT_s)}$ Ⓑ

---

9)



$std(x) = \sigma$, $\sigma^2 = 2 \rightarrow \sigma = \sqrt{2} = \mu$

$$z = \frac{x - \mu}{\sigma} \xrightarrow{(\mu = x)} z = \frac{\mu - \mu}{\sigma} = 0$$

$P(x > std\,x) = P(x > \mu) = P(z > 0) = 0.5$

10) $y = \sqrt[3]{x} \Rightarrow \boxed{x = y^3}$ $\quad$ $p.d.f = x = y^3$

$$\frac{dx}{dy} = 3y^2$$

$$f_y(y) = f_x(T^{-1}(y)) \cdot \left| \frac{d}{dy} T^{-1}(y) \right|$$

$= y^3 \cdot 3y^2 \Rightarrow \boxed{3y^5}$ Ⓓ

مكرر ابو دية
Final 2025

---

11) $C_{xy} = R_{xy} - \bar{x}\,\bar{y} \Rightarrow$ jointly wide-s

$C_{xy} = 2 + \cancel{\bar{x}}\cancel{\bar{y}} - \cancel{\bar{x}}\cancel{\bar{y}} \Rightarrow \boxed{C_{xy} = 2}$ Ⓐ

---

12) A ✓ $\quad$ Ⓑ

B ✗

C ✓

D ✓

---

13) $\boxed{R_{xy} = 0}$ $\quad$ $\bar{x}\,\bar{y} = 25$ ① $\quad$ $\bar{x} = ??$

$\bar{x} + \bar{y} = 10$ ②

$\boxed{\bar{y} = 10 - \bar{x}} \Rightarrow \boxed{10\bar{x} - \bar{x}^2 = 25} \Rightarrow \bar{x}^2 - 10\bar{x} + 25 = 0$

$(\bar{x} - 5)(\bar{x} - 5) = 0$

$\boxed{\bar{x} = 5}$ Ⓒ

---

14) $\boxed{\mathrm{corr}(2x, y) = 2\,\mathrm{corr}(x, y)}$ $\quad$ $xy = x^2 - x \cdot E(x)$

$\downarrow$

$= 2 * E(xy)$ $\quad$ $E(xy) = E(x^2) - (E(x))^2$

$\boxed{2 * 3 = 6}$ Ⓓ $\quad$ $\boxed{E(xy) = 3}$

لجنة السنافر